

An Efficient Synthetic Data Generation Algorithm to Improve Efficacy of Deep Learning Models of Medical Images

Sai Kiran Maryada (✉ sai.maryada@ou.edu)

University of Oklahoma <https://orcid.org/0000-0003-1982-1012>

William Lee Booker

Babel Analytics

Gopichandh Danala (✉ danala@ou.edu)

University of Oklahoma

Meredith Jones

University of Oklahoma

Sanjana Mudduluru

University of Oklahoma

Huong Pham

University of Oklahoma

Dean F. Hougen

University of Oklahoma

Bin Zheng

University of Oklahoma

Research Article

Keywords: Generating synthetic image data, deep transfer learning, retinal fundus images, retinal fundus disease detection and classification, convolutional neural networks

DOI: <https://doi.org/10.21203/rs.3.rs-2416807/v1>

License: © ⓘ This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

An Efficient Synthetic Data Generation Algorithm to Improve Efficacy of Deep Learning Models of Medical Images

Sai Kiran Maryada¹, William Lee Booker², Gopichandh Danala³, Meredith Jones⁴, Sanjana Mudduluru¹, Huong Pham³, Dean F. Hougen¹, Bin Zheng³

¹School of Computer Science, University of Oklahoma, Norman, OK 73019, USA

²Babel Analytics, Tulsa, OK 74137, USA

³School of Electrical and Computer Engineering, University of Oklahoma, Norman, OK 73019,
USA

⁴Stephenson School of Biomedical Engineering, University of Oklahoma, Norman, OK 73019, USA.

Abstract:

Deep Convolutional Neural Networks (DCNNs) have emerged as powerful components of computer-aided systems to detect and classify diseases in medical images. In order to robustly achieve high performance, a large and diverse dataset with annotated images is required to train or fine-tune DCNNs. However, such datasets are often very difficult and/or expensive to obtain due to demands on clinicians' time and inter-reader variability. To address these challenges and increase training dataset size, we propose an efficient multi-stage algorithm to generate synthetic medical image data by extracting annotated diseased regions and randomly projecting them onto disease-free images. To test the feasibility of this new algorithm, the publicly available Indian Diabetic Retinopathy Image Dataset (IDRiD) is used. This dataset is comprised of the annotated fundus images acquired from 81 patients with two categories of diseases. Among them, 54 and 27 images are used for training and testing images, respectively. Using the proposed algorithm, we generate synthetic data by inserting extracted diseased lesions onto another set of 60 disease-free images, which results in 7,902 and 6,786 images for the two categories of diseases, respectively. Three transfer-learning-based DCNN models (VGG16, ResNet50, and Inception-v3) are trained using original IDRiD images and synthetic dataset, respectively. When applied to the same test images, the model trained with the synthetic dataset outperformed the model trained using the original IDRiD dataset by 7.4% in disease classification. These results indicate that this simple algorithm can generate diverse and useful synthetic lesion data to improve performance of DCNN models.

Keywords: Generating synthetic image data, deep transfer learning, retinal fundus images, retinal fundus disease detection and classification.

1. Introduction

Medical images are widely used to detect and diagnose diseases. As a result, imaging data accounts for approximately 90% of all clinically diagnostic data generated in healthcare systems [1]. However, reading and interpreting medical images by clinicians is a difficult and time-consuming task which results in large intra- and inter-observer variability [2]. To assist clinicians to more accurately read and diagnose medical images with less variability,

many computer-aided detection and diagnosis (CAD) schemes for medical images have been developed in the last two decades for a variety of applications including identifying quantitative image markers, detecting diseases, classifying disease types or severities, and predicting disease prognosis or response to treatment [3-9]. Despite significant progress in developing CAD schemes of medical images, accurate segmentation of medical images and identification or selection of effectively handcrafted image features to train traditional machine learning classifiers remains difficult and not robust. Thus, the clinical utility and the value added to clinical practice by using CAD schemes is still limited [10].

To overcome the limitations in developing CAD schemes, incorporating deep-learning or deep-transfer-learning models based on deep convolutional neural networks (DCNNs) into CAD systems for medical images has been attracting broad research interest in recent years [11–13]. The advantages of using DCNN models over traditional machine learning classifiers based on handcrafted features to develop CAD systems for medical images have been demonstrated empirically [14, 15]. However, although using deep-learning models bypasses the need for image segmentation and handcrafted feature computation, it often requires much larger and more diverse image datasets to train deep-learning models or fine-tune deep-transfer-learning models in order to reduce bias and achieve high levels of robustness. For example, one study developed and validated a deep-learning model to detect diabetic retinopathy using retinal fundus photographs. The study reported that using a deep learning model yielded very high detection accuracy with an area under the receiver operating characteristic curve or AUC of 0.991 with a 95% confidence interval of 0.988 – 0.993 [16]. This model was trained using 128,175 retinal images that were annotated by 54 ophthalmologists or ophthalmology senior residents.

However, in most medical imaging research tasks to develop new CAD schemes using DCNN models, the size of the image dataset is much smaller or limited due to the difficulty of the task and a lack of resources to identify and retrieve clinical images from hospitals and ask clinicians to label or annotate the diseased area or patterns in the retrieved medical images. In addition, established small datasets are often also imbalanced between *positive* (diseased) and *negative* (healthy) cases; the number of positive cases is usually much smaller than the number of negative cases. One study reported that segmenting and labeling diseased regions usually consumes around 15 minutes per image, making it impractical for clinicians (i.e., radiologists) in their very limited research or academic time to review and annotate large numbers of images [17]. This results in a scarcity of annotated medical image data for training and testing DCNN models in the medical imaging research field.

Currently, in order to apply DCNN models the medical imaging field, two approaches are commonly used. The first one uses *data augmentation* techniques to increase the number of training data items and directly mitigate the paucity of annotated data. The second one is to apply deep transfer learning to indirectly mitigate or reduce the effect of the labeled data scarcity. *Transfer learning* focuses on retaining knowledge gained while solving a problem from one domain and applying it to a different domain. Several research studies [16, 18, 19] show that fine-tuning DCCN models originally trained on a large and diverse dataset of natural images (i.e., ImageNet [20]) using small datasets of medical images helped improve accuracy and decreased time for convergence of model training or fine-tuning. However, since real clinical images are quite heterogeneous not only in the diseased regions or lesions, but also in

healthy or normal tissues patterns or background, these two approaches can only partially reduce the impact of small image datasets because they cannot increase the diversity of medical image cases in the data.

Thus, in order to find a better solution, researchers have also investigated whether adding synthesized data [21] generated from real medical image data can help overcome the above-mentioned issues and thus develop more robust DCNN models. One recent study [22] shows that the synthetic images generated by a generative adversarial network (GAN) were not recognized as synthetic by radiologists and that using these GAN-generated synthetic images helped increase the performance of a DCNN model in classifying liver lesions. Thus, using GANs is shown to be a promising solution to generate more synthetic image data across many application domains including medical imaging. Recent studies have reported that GANs have been applied to generate synthetic images of skin lesions [23], colon mucosa [24], and chest X-ray images [25].

However, GANs are very complex machine-learning models; how they can efficiently generate large numbers of diverse or heterogeneous medical images, particularly depicting lesions or diseases against a wide variety of normal tissue backgrounds, has not been well investigated. In this study, we propose a new and more computationally efficient algorithm without GANs to generate synthetic image data on a diverse image base. The new algorithm extracts the annotated diseased region or lesion blobs, randomly redistributes these lesion blobs and projects them onto negative (healthy) images to optimally achieve seamless insertion. Unlike a GAN approach that is limited by a small number of annotated positive lesions, our new method can include more diverse negative images that do not need expert annotations. In clinical practice the number of normal (disease-free) images are the majority and are easy to acquire for research purposes. Thus, using this new algorithm can produce diverse synthetic image data generated using negative images with a broad diversity or heterogeneity of normal tissue patterns, which we hypothesize can help better train DCNN models and improve model performance. Therefore, the objective of this study is to test or validate the feasibility of our hypothesis by comparing three deep transfer learning models (VGG-16 [26], ResNet-50 [27], and Inception V3 [28]) trained using a small set of original images and a large set of the synthetic images, respectively. Recently, these three deep learning models have been widely used in developing new CAD schemes of medical images with promising results.

To report this study in detail, the rest of this paper is organized as following. In Section 2, we present study materials and methods. We start this section by describing the baseline dataset and then propose the synthetic data generation algorithm. We also explain the different CNN architectures used to verify the effectiveness of the synthetic data generation algorithm. Section 3 presents experiment and data analysis results including plots and figures. It also includes a summarization table of relevant performance metrics comparing models developed using both the original image data and our synthetic data. Section 4 discusses the various synthetic data generation methods, the advantages and potential of applying our method to other medical images, as well as limitations of our algorithm and this study.

2. Materials and Methods

2.1. Dataset and Image Preprocessing

For this study we first used the Indian Diabetic Retinopathy Image Dataset (IDRiD), a publicly available dataset of retinal fundus images [29]. The dataset consists of retinal images of 81 patients along with corresponding annotated binary masks for three types of lesions: Hemorrhages (HE), Hard Exudates (EX), and Soft Exudates (SE). In this dataset, all three types of diseased lesions have been annotated in the images by clinicians. This dataset is randomly divided into two independent training and testing subsets with 54 retinal fundus images selected for training and 27 images for testing.

In this dataset, each patient falls into one of two categories. In *Category 1*, an image contains all three types of lesions, while in *Category 2*, an image contains only HE and EX lesions. In the training subset, 26 and 28 images belong to Categories 1 and 2, respectively. In the testing subset, 14 and 13 images are assigned to Categories 1 and 2, respectively. Each case includes one original retinal fundus image and 3 or 2 annotated mask images, one for each type of disease in Categories 1 and 2, respectively. Figure 1 shows an example image in Category 1 and three corresponding masks of HE, EX, and SE lesions. Figure 2 shows another example image belonging to Category 2 along with the two associated masks of HE and EX lesions. All labeled mask images are pseudo binary images without lesion density information.

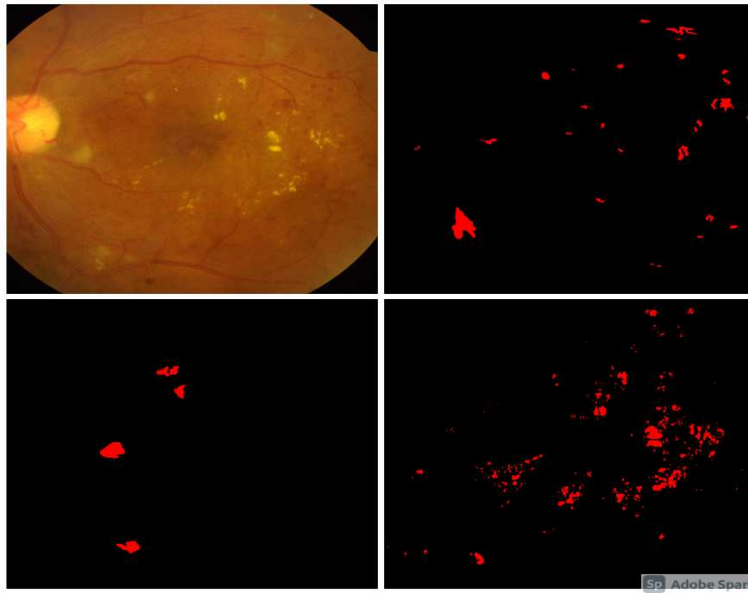


Figure 1: Original retinal image (top left) with three annotated masks for three types of lesions (top right-clockwise: HE, EX, and SE).

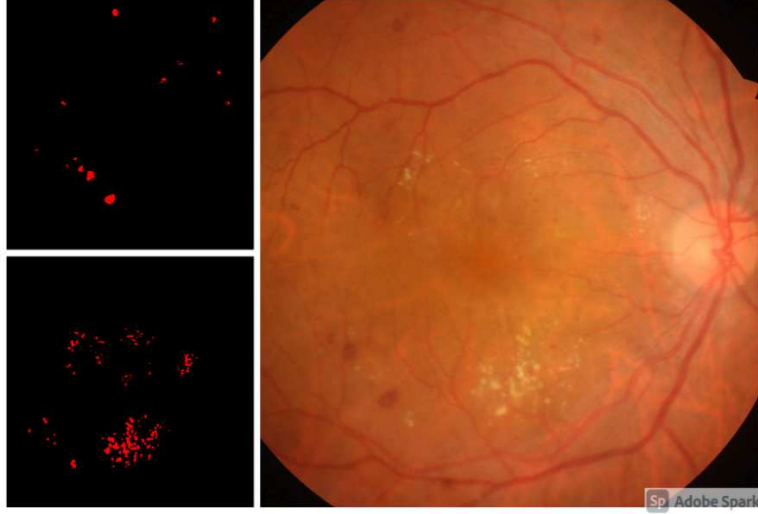


Figure 2: Original retinal image (right side) with two annotated masks for two types of lesions (top left: HE; bottom left: EX).

To prepare the images for use in the study, we applied following image preprocessing steps. First, edge detection was performed horizontally on each retinal fundus image to find the maximum width of the fundus within the image. Second, the image was cropped to this width to remove excess background from the image. Third, a black (background) padding method was applied vertically to convert the image into a square. Last, each image was resized to a square region of interest (ROI) of 225×225 pixels, which matches the input image size of the VGG-16 and ResNet-50 DCNN models. Figure 3 shows an example of this image preprocessing result. The dark background regions on the left and right side are removed and two padding strips are added on the top and bottom of the images to convert the image into a square ROI with 225×225 pixels. Similarly, we have added extra padding to generate another set of images with square ROI of size 299×299 pixels, which is the input image size for inception V3.

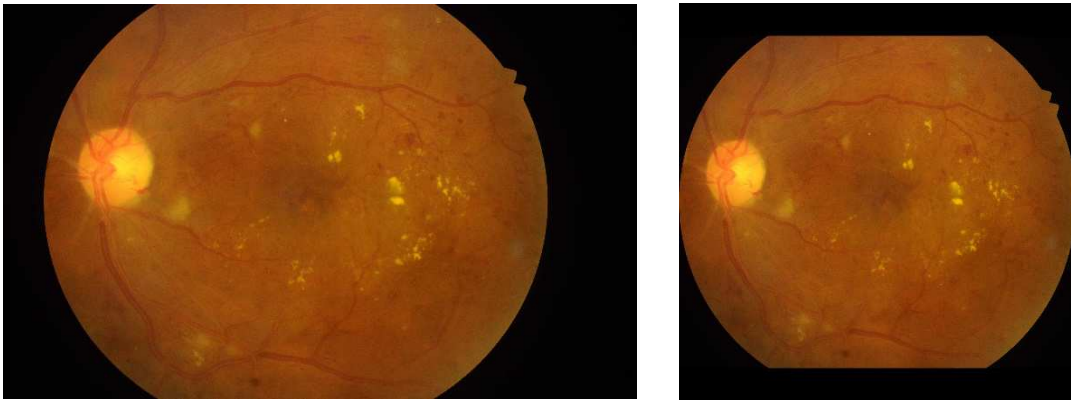


Figure 3: Original retinal image (left side); image after pre-processing (right side).

2.2. Synthetic Data Generation Algorithm

To generate synthetic images for this study, we acquired another dataset of 60 healthy (or normal) retinal fundus images, which were provided by an ophthalmologist in our local clinic as a base to generate images with synthetically added diseased patterns or lesions. Then we designed and applied the following new synthetic data generation algorithm to extract lesion blobs, randomize their distribution to create new lesions, and project the synthetic lesions onto these healthy retinal fundus images. The algorithm includes the following six major steps.

1. Because there are three separate disease masks of HE, EX, and SE lesions in Category 1 images (as shown in Figure 1) and two separate disease masks of HE and EX lesions in Category 2 images (as shown in Figure 2) in the IDRiD dataset, we created a combined mask image that contains either three types of lesions for the Category 1 disease case or two types of lesions for the Category 2 disease case in the training subset by using element-wise addition, which means inserting the lesions extracted in the original masks into a uniform “black” mask. The top row of Figure 4 shows one combined mask that contains three types of lesions extracted from one Category 1 disease case. As a result, this step creates 54 combined masks in which 26 represent masks of Category 1 cases and 28 represent Category 2 cases.
2. After creating the combined mask image of each case, we count the number of lesion blobs (N) in each mask image and label them from 1 to N . After examination of whole dataset, we noticed that the counted number of lesion blobs (N) varies from a minimum of 20 to more than 100 in different mask images. Depending on the number of lesion blobs (N) associated with each case, we randomly select 2 to N blobs from each originally combined mask to generate 20 new masks. Thus, in this step, we generate 1,080 new mask images (54 combined masks $\times$ 20 new masks per combined mask) that each contain a random number of lesion blobs. Figure 4 shows an example of one originally combined lesion mask and 20 new lesion masks generated from this combined lesion mask.
3. We use element-wise multiplication to identify the lesion type of each blob in the mask randomly generated in Step 2. The results are saved in a database that records the location of each blob and the associated lesion type (HE, EX, or SE). Since the masks generated in Step 2 are pseudo-binary mask images, in this step we map them into new mask images that contain real lesion blobs. For this purpose, we simply project the pseudo blobs in the masks generated in Step 2 back to the original retinal fundus images to extract real lesion blobs. As a result, the new mask images contain real blobs that contain lesion density information.
4. Each real lesion blob mask obtained in Step 3 is flipped both horizontally and vertically. Thus, one real lesion mask becomes three masks (original, horizontally flipped, and vertically flipped) with the same set of real lesion blobs located in different positions and orientations. As a result, we generate a total of 3,240 real lesion masks ($1,080$ generated mask images $\times$ 3 transformation options). Due to the random selection process, the final real lesion blob masks can be divided into seven categories containing lesions of (1) HE + SE + EX, (2) HE + SE, (3) EX + SE, (4) HE + EX, (5) HE, (6) SE, (7) EX. Among these masks, most of them contain EX + SE and HE + SE + EX categories; the number of masks in the other five categories are much smaller. In order to overcome this data imbalance issue, we performed multiple angular rotations to increase the numbers of masks in the five categories to generate approximately comparable numbers. Table 1 shows the resulting number of the final

synthetic lesion blob masks in each of the seven categories. While we did not need to generate datasets for all seven categories for the experiments presented herein, we did so to demonstrate the ability of the algorithm to generate data in categories beyond those that appear in the original dataset.

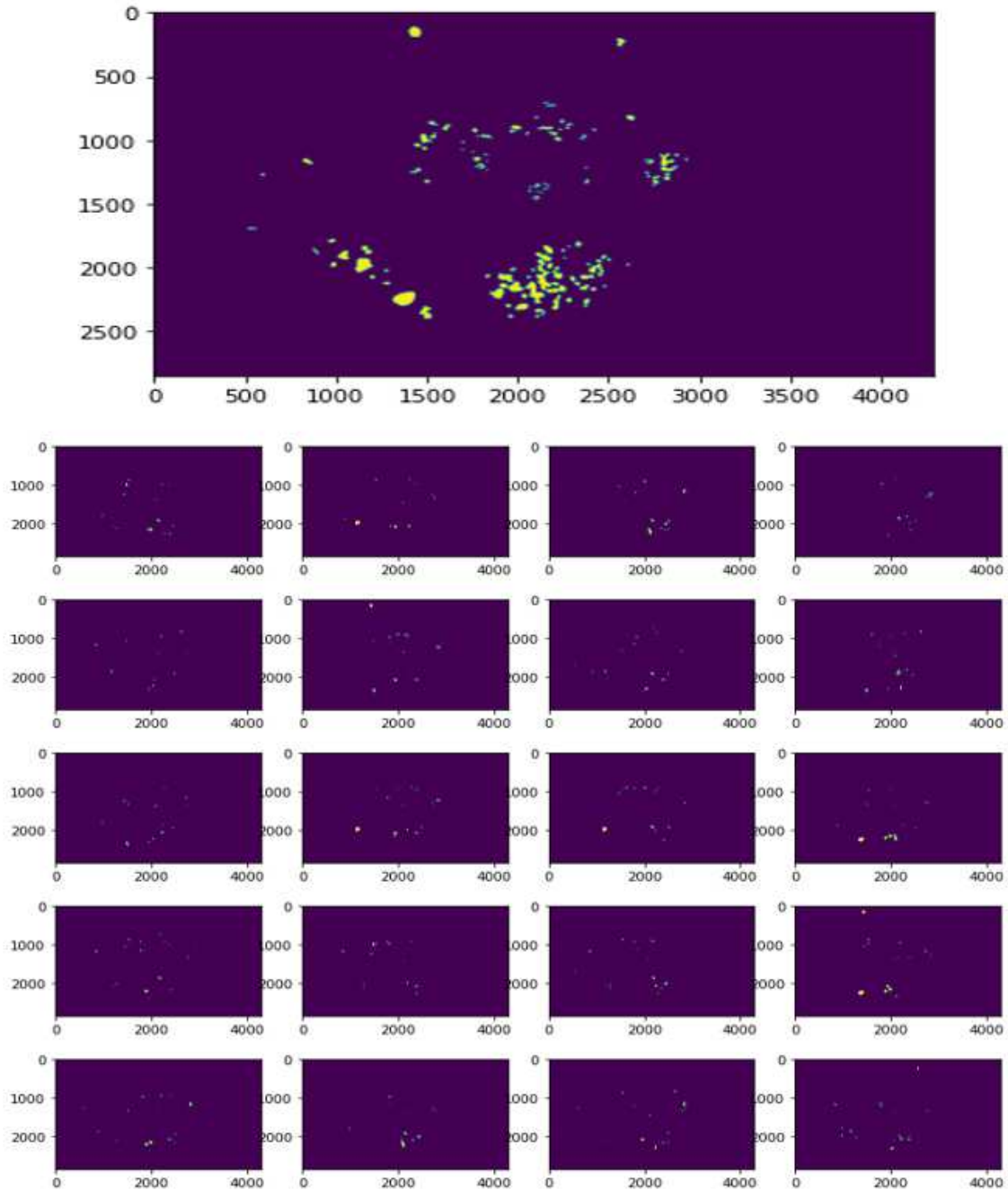


Figure 4: The image on top shows one originally combined mask image that consists of all 3 types of lesions (Category 1) and images below on the grid show 20 new mask images generated based on the top combined mask image with random distributions of lesion blobs.

Table 1: Numbers of synthetic real lesion blob masks generated in seven categories of lesion distributions

Category	Lesion composition	Number of images
1	HE, SE, and EX	7092
2	HE and SE	6786
3	EX and SE	6939
4	HE and EX	7902
5	HE	7641
6	SE	6780
7	EX	6780

- Finally, we insert and project each of the lesion blob masks generated in Step 4 onto a randomly chosen healthy retinal fundus image. In the projection, the overlapping pixels between the healthy retinal fundus image and the synthetic lesion blob mask are taken from the synthetic lesion blob mask, whereas the remaining pixels are extracted from the healthy fundus image. As a result, we generated 7,092 synthetic images of Category 1 and 6,786 synthetic images of Category 2, as shown in Table 1, which were then used to train deep learning model as described below.

In summary, Figure 5 illustrates the step-by-step work flow of the proposed synthetic image data generation algorithm in which the figure shows an original retinal fundus image (Figure 5a) and corresponding lesion mask (Figure 5b) these are taken from IDRiD dataset, a randomly generated lesion mask (Figure 5c), a lesion blob image that is obtained after projecting the mask with the corresponding original image (Figure 5d), a sample flip (horizontal, Figure 5e), a random angular orientation (Figure 5f), and a lesion blob mask insertion to a healthy retinal fundus image to generate the final synthetic image (Figure 5g). Figure 6 demonstrates nine synthetic images on the right side and one magnified view of one synthetic image with three clusters of blobs annotated on the left side.

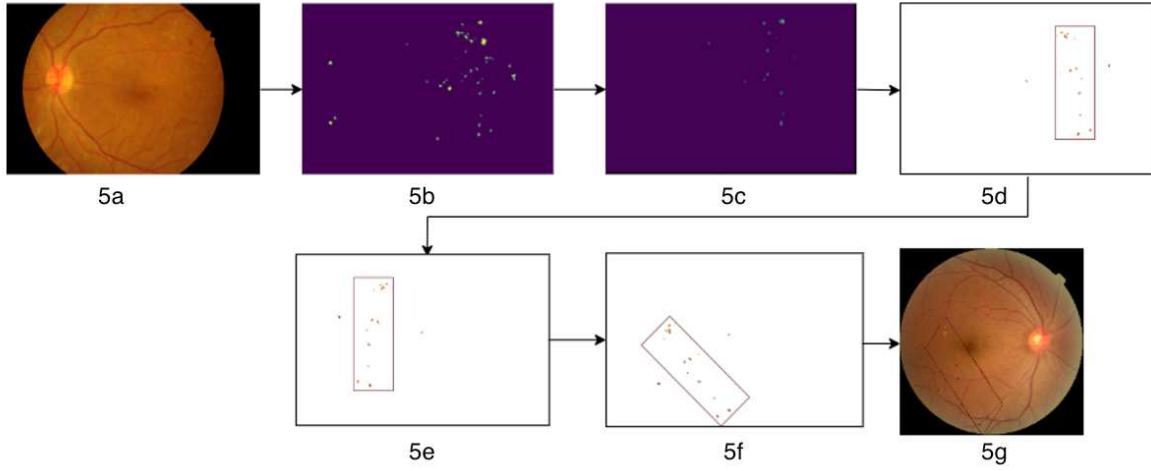


Figure 5: A detailed step-by-step illustration of the proposed algorithm.

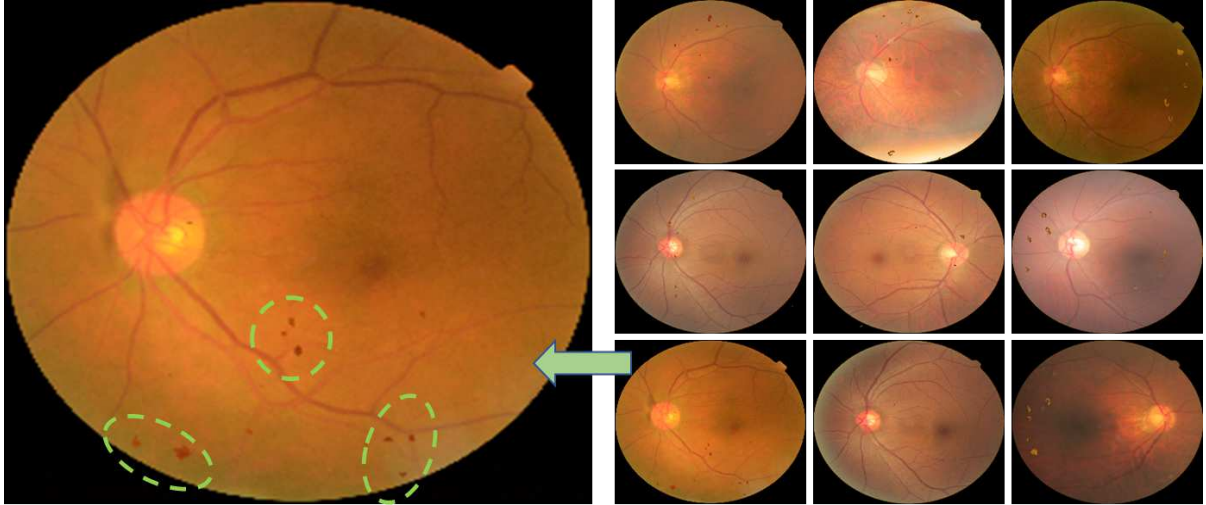


Figure 6: Synthetic data after projecting lesions onto healthy images (right side); magnified view of one sample case with some lesions annotated (left side).

2.3. Experiments

In this study, we built six deep-transfer-learning DCNN models with three different architectures aiming to classify retinal fundus images into two different disease categories. Although many DCNN models have been developed and used as transfer learning models in medical imaging informatics field, we chose three popular DCNN models namely, VGG16, ResNet-50, and Inception-v3 architectures [30].

Two VGG16-network-based deep transfer learning models were trained using the original image data provided in the IDRiD dataset and the synthetic image data generated by the new algorithm developed in this study, respectively. Figure 7 shows the modified VGG-16 architecture. Specifically, Model-1 was trained using 26 Category 1 images and

28 Category 2 images acquired from 54 original IDRiD images, while Model-2 was trained using 7,092 Category 1 synthetic images and 6,786 Category 2 synthetic images. Since the VGG16 architecture is pre-trained on the ImageNet database and accepts input images with size 225×225 pixels, all retinal fundus images were resized to be 225×225 pixels. Next, two ResNet-50 network based deep learning models and two Inception-v3 network based deep learning models were trained using original data and synthetic data, respectively. Figure 8 and Figure 9 show the deep learning architecture of the modified ResNet-50 network and the modified Inception-v3 network, respectively.

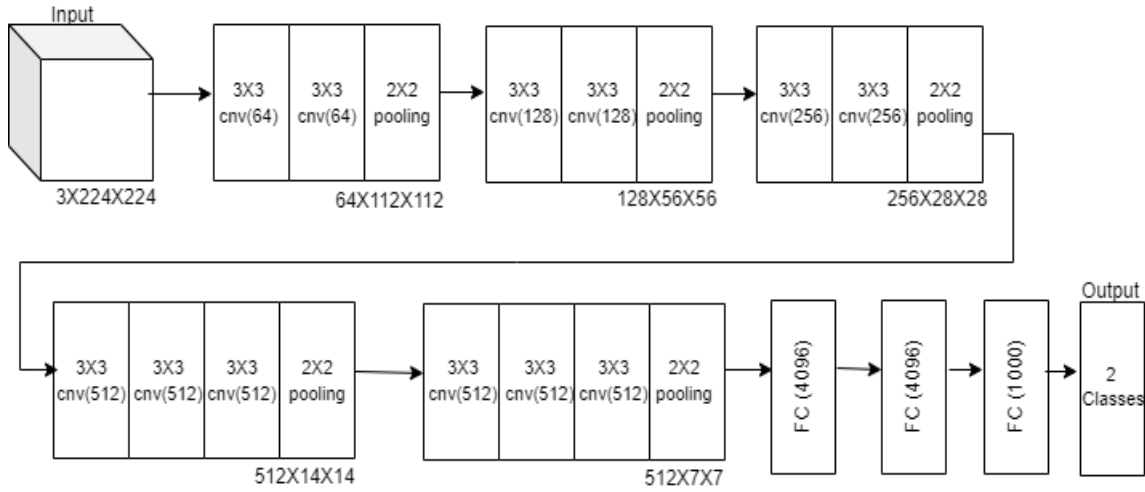


Figure 7: Illustration of the modified VGG-16 network.

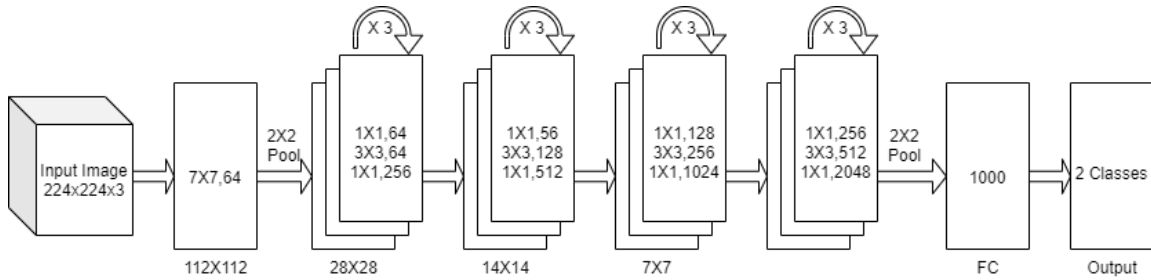


Figure 8: Illustration of the modified ResNet-50 network.

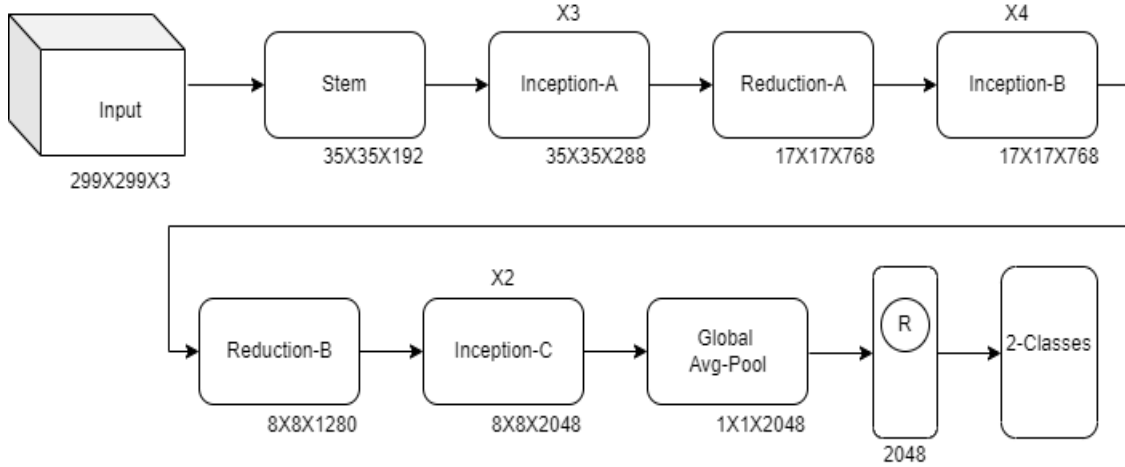


Figure 9: Illustration of the modified Inception-v3 network.

Using either real or synthetic retinal fundus images, the VGG-16, ResNet-500, and Inception-v3 based DCNN models were fine-tuned to classify images as Category 1 or Category 2 retinal diseases. In the fine-tuning process, the Adam optimizer is adapted to use a variable learning rate that starts from 0.006 and exponentially decays by a factor of 0.05 for every three epochs. To minimize or reduce overfitting risk, models are trained using 25 epochs based on the cross-entropy loss of the Adam optimizer. Then, the last SoftMax layer of each model is modified and changed to one output neuron with sigmoid activation that achieve the goal of classifying two categories of retinal fundus images in this study.

After fine-tuning and optimizing each DCNN model, each model was applied to the same independent test dataset of 27 images provided in the IDRiD dataset. Since the last SoftMax layer has two output nodes that generate two probability scores indicating the likelihood of a test image belonging to two disease categories, the test image is assigned to the category that has the higher probability score. From the test results, we calculated true positive (TP), false negative (FN), true negative (TN), and false positive (FP) values to generate a confusion matrix. Then, from the confusion matrix, we evaluate model classification performance by computing four commonly used evaluation indices: accuracy, precision, recall, and F1 score, using Equations (1) – (4).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

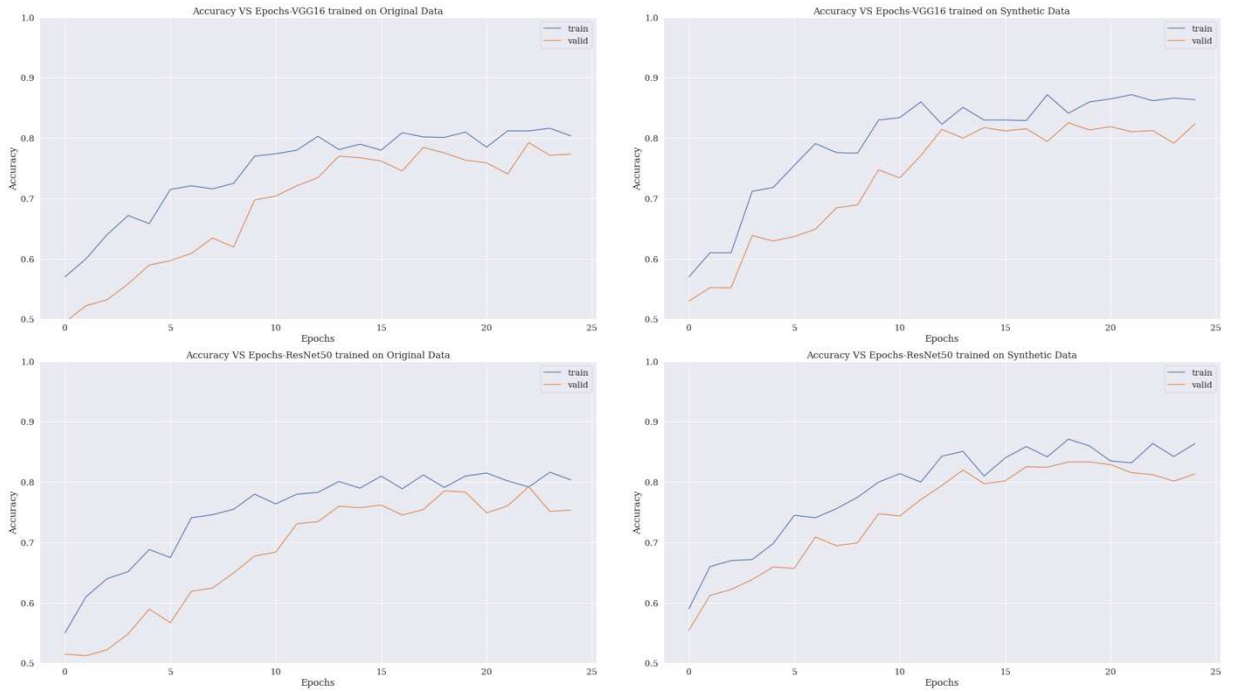
$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

Then, these classification performance indices of the different DCNN models separately trained using the original IDRiD images and the synthetic images were tabulated and compared.

3. Results

Figure 10 shows six diagrams that plot curves of disease classification accuracy over 25 training epochs of the three deep transfer learning models (VGG16, ResNet-50, and Inception-v3) trained using the original IDRiD images and the synthetic images, respectively. In training or fine-tuning each model, the training dataset was divided into two subsets in which 80% of the image data was used to train the model and 20% of image data was used to validate the performance of the model. The results of the two trend curves generated by the training and validation data indicate that the classification accuracy approaches a plateau and gradually saturates once epochs exceed 20. Thus, in this study we chose 25 training epochs for all six models as shown in Figure 10.



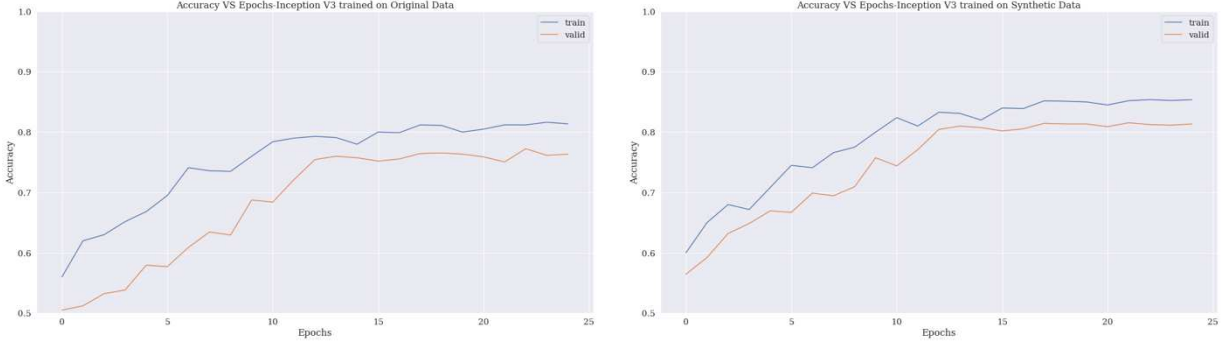
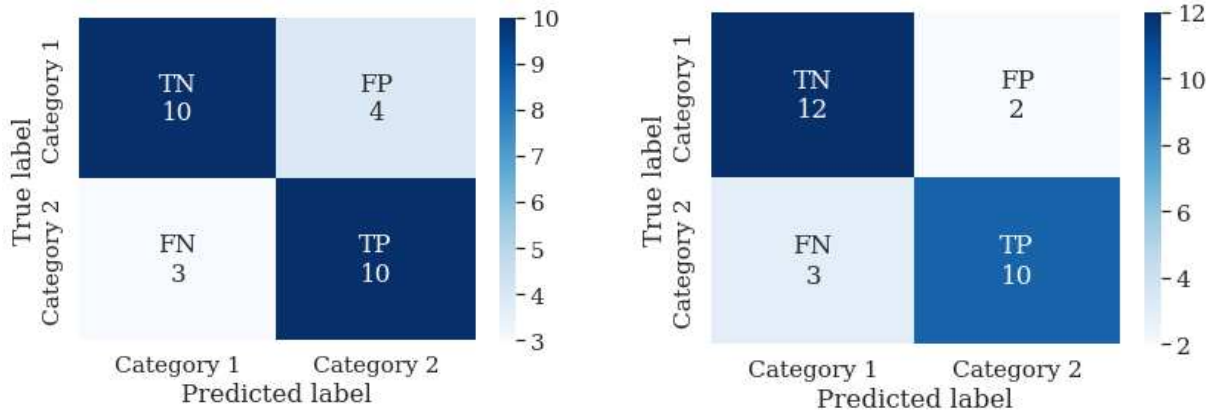


Figure 10: Illustration of training and validation accuracy curves of three deep transfer learning models, VGG16 (top), ResNet-50 (middle), and Inception-V3 (bottom) models trained using the original IDRiD images (left) and the synthetic images (right)

The testing dataset includes 14 images in Category 1 that includes Hemorrhages, Hard Exudates, and Soft Exudates, and 13 images in Category 2 that includes Hemorrhages and Hard Exudates. Figure 11 demonstrates three sets of two confusion matrices generated by applying three transfer learning DCNN models (VGG-16, ResNet-50, and Inception-v3) trained using the original IDRiD images and the synthetic images to the same testing dataset of 27 images to classify between Category 1 and Category 2 diseases, respectively. Based on these confusion matrices, Table 2 shows and compares four evaluation indices of classification performance generated by three deep transfer learning models trained using the original IDRiD images and the synthetic images. The results show that all three transfer learning models (VGG-16, ResNet-50, and Inception-v3) yield an overall classification accuracy of 81.5% (22/27), which is 7.4% higher than the classification accuracy of 74.1% (20/27) using all the three models (VGG-16, ResNet-50, and Inception-v3) trained on the original IDRiD images.



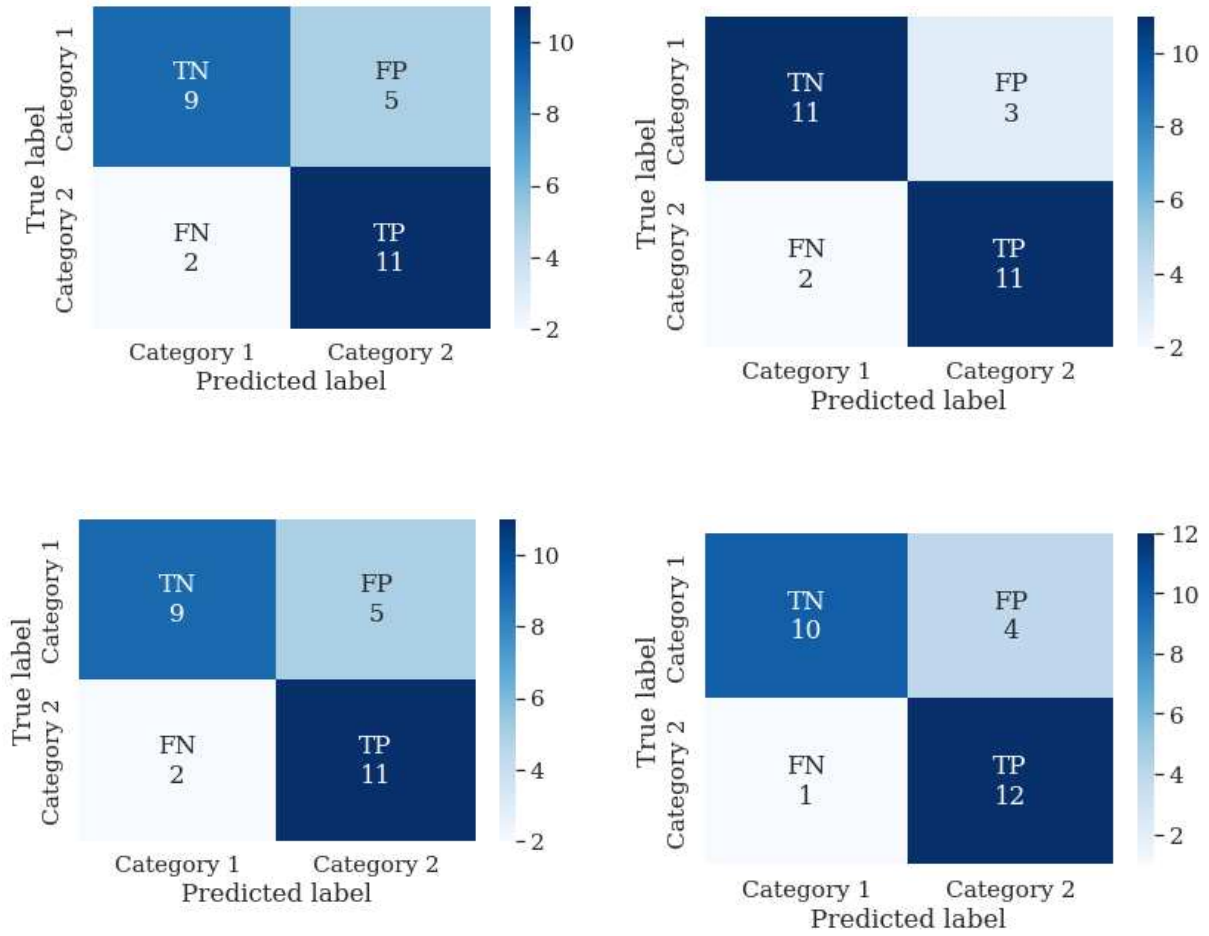


Figure 11: Six confusion matrices generated by VGG16 (top), ResNet-50 (middle) and Inception-v3 (bottom) trained using original IDRiD images (left) and synthetic images (right).

Table 2: Comparison of various performance metrics while using original and synthetic data

Model	Training Images	Accuracy	Precision	Recall	F1 Score
VGG-16	Original	0.741	0.714	0.769	0.741
	Synthetic	0.815	0.833	0.769	0.800
ResNet-50	Original	0.741	0.688	0.846	0.759
	Synthetic	0.815	0.786	0.846	0.815
Inception-v3	Original	0.741	0.688	0.846	0.759
	Synthetic	0.815	0.750	0.923	0.828

Although all three models (VGG-16, ResNet-50, and Inception-v3) yield the same classification accuracy, the distribution of other three evaluation indices (precision, recall, and F1 score) are different among the three models.

The transfer learning models fine-tuned using the original IDRiD images yield substantially higher precision as the models fine-tuned using the synthetic images. The results indicate that the models fine-tuned using synthetic images achieve higher sensitivity and generally higher false positive rates. Combining precision and recall indices, the models fine-tuned using synthetic images yield substantially higher F1 scores in this study. For example, by comparing two Inception-v3 transfer learning models, testing results indicate that using synthetic data to fine-tune the model increases F1 score by 6.9% (from 0.759 to 0.828).

4. Discussion

Due to the difficulty of acquiring large datasets of the well-annotated medical images for medical imaging research, generating the reliable and diverse synthetic images plays an important role to improve efficacy of either building new deep learning models or fine-tuning the existing deep learning models for medical imaging applications. Thus, there is broad interest in medical imaging research field to develop more efficient and robust algorithms to generate synthetic image data with high clinical relevance. In this study, we propose and demonstrate a new algorithm to generate synthetic retinal fundus images embedded with different types of diseased lesions. The new synthetic image data generation method has several unique characteristics, and the study results also generate several interesting observations.

First, current existing synthetic image data generation methods or algorithms include a Monte Carlo method based on repeated random sampling and statistical analysis of the results, an unsupervised Variational Auto-Encoder (VAE) algorithm that learns the distribution of the original image dataset and generate synthetic image data via double transformation using an encoded-decoded architecture, and a Generative Adversarial Network (GAN) that uses two neural networks working together to generate fake yet realistic data points. These methods are quite complicated or difficult to design, train, and implement, and are computationally expensive because large numbers of algorithm or network parameters need to be chosen and optimized. However, our new method is much simpler and computationally efficient. It uses a multi-stage approach to directly extract positive lesions, randomize distribution of lesion blobs, and then inserts or projects the positive lesion blobs in randomly selected locations of negative images. In the medical imaging field collecting large numbers of negative images is much easier than collecting positive images with manually annotated diseased regions or lesions. This study demonstrates that by using this algorithm, we can significantly expand our dataset size from 54 original positive images and 60 negative or normal images to 14,994 synthetic images (for Categories 1 and 2 diseases as shown in Table 1).

Second, although the original 54 images are divided into only two categories, using this new algorithm, we can generate all varieties of patient data (Categories 1-7) as shown in Table 1. In this study, the algorithm generates 49,920 synthetic images in 7 categories. Please note that despite the small image base of 54 positive images and 60 negative images, and large number of synthetic images, all algorithm-generated synthetic images have different positive lesion blob combinations, and the lesion blobs randomly distribute in different locations with different orientations, which increases diversity of training samples by avoiding or minimizing redundancy of the generated synthetic image data.

Third, we fine-tuned three sets of DCNN models based VGG16, ResNet-50 and Inception-V3 model. Each DCNN set includes two models. Model-1 is trained using original IDRiD images and Model-2 is trained using the algorithm-generated synthetic image data. Two models are then applied to the same testing images. The experimental results show that Model-2 yields higher classification accuracy than Model-1. Specifically, among 27 testing cases, Model-1 correctly classifies 20 cases, while using Model-2 correctly classifies 22 cases with a classification accuracy increase by 7.4%. We recognize that this accuracy improvement is observed on a quite small testing image dataset of 27 images. The two models should be further tested using much large image datasets in the future. However, the results still demonstrate the feasibility and advantages of using new simple algorithm to generate synthetic images to substantially increase size of training dataset and potentially increase model classification performance on the new independent testing images.

Fourth, we also observe that classification performance as measured by 4 evaluation indices (as shown in Table 2) is independent from three deep learning DCNN models (VGG-16, ResNet-50 and Inception-V3). It is promising that all three types of different DCNN models that are fine-tuned using synthetic images achieve substantially higher F1 scores ranging from 5.6% (ResNet-50) to 6.9% (Inception-v3) as shown in Table 2. This observation indicates the robustness or scientific rigor of applying this new simple multi-step algorithm to generate synthetic images that can effectively help train or fine-tune different DCNN models.

Finally, although study results are promising, we also recognize that seamless insertion of lesions or other abnormality regions onto negative images using this simple algorithm has restrictions or limitations, which include that (1) the lesions must have clear boundary contours so that the lesions can be easily segmented or extracted, (2) the normal tissue background should also be relatively uniform. Retinal fundus images meet these two restrictions. Some other medical images (i.e., liver tumors) can also meet these restrictions. Nonetheless, if lesions have fuzzy or irregular boundary embedded under heterogenous tissue background (i.e., breast tumors depicting on mammograms), this algorithm will not work “as is” and modifications will be needed. Despite such limitations, developing this new simple algorithm to generate synthetic images still has its higher clinical application potential or impact at least for retinal fundus images that are acquired using a low-cost image examination method and thus widely used in clinics to screen, detect, and diagnose many common human diseases including a variety of eye diseases and diabetes. We will further test and validate this new algorithm and apply it to develop more accurate and robust deep learning models for different medical applications in the future.

Acknowledgement:

This work is supported in part by the National Cancer Institute, National Institutes of Health, under Contract No. 75N91020C00047, the National Institute of General Medical Sciences, National Institutes of Health, under Grant No. P20GM135009, and the Oklahoma Center for the Advancement of Science and Technology through the award for project number IP21-024.

Conflict of Interest: Sai Kiran Maryada, William Lee Booker, Gopichandh Danala, Meredith Jones, Sanjana Mudduluru, Huong Pham, Dean F. Hougen, Bin Zheng declare they have no conflict of interest.

References

- [1] R. L. Van Metter, J. Beutel, H. L. Kundel, *Handbook of Medical Imaging*. SPIE, 2000.
- [2] B. Zhao, Y. Tan, D.J. Bell, *et al.*, “Exploring intra- and inter-reader variability in uni-dimensional, bi-dimensional, and volumetric measurements of solid tumors on CT scans reconstructed at different slice intervals,” *European Journal of Radiology*, 2013; 82:959–968,
- [3] G. Danala, S. K. R. Maryada, M. Heidari, *et al.*, “A new interactive visual-aided decision-making supporting tool to predict severity of acute ischemic stroke,” *Proc SPIE*, 2020; 113171V.
- [4] M. Heidari, S. Lakshmivarahan, S. Mirniaharikandehi, *et al.*, “Applying a Random Projection Algorithm to Optimize Machine Learning Model for Breast Lesion Classification,” *IEEE Transactions on Biomedical Engineering*, 2021; 68(9):2764–2775.
- [5] S. K. R. Maryada, *et al.*, “Applying a novel two-stage deep-learning model to improve accuracy in detecting retinal fundus images,” *Proc SPIE*, 2022; 120330K.
- [6] G. Danala, *et al.*, “Applying quantitative radiographic image markers to predict clinical complications after aneurysmal subarachnoid hemorrhage: A pilot study,” *Annals of Biomedical Engineering*, 2022; 50(4):413–425.
- [7] T. Gai, T. Thai, M. Jones, *et al.*, “Applying a radiomics-based CAD scheme to classify between malignant and benign pancreatic tumors using CT images,” *Journal of X-ray Science and Technology*, 2022; 30(2):377-388.
- [8] G. Danala, B. Ray, M. Desai, *et al.*, “Developing new quantitative CT image markers to predict prognosis of acute ischemic stroke patients,” *Journal of X-Ray Science and Technology*, 2022; 30(3):459–475.
- [9] G. Danala, *et al.*, “Developing interactive computer-aided detection tools to support translational clinical research,” *Proc SPIE*, 2022; 1203503.
- [10] R.M. Nishikawa, D. Gur, “CADE for early detection of breast cancer—current status and why we need to continue to explore new approaches,” *Academic Radiology*, 2014; 21(10):1320–1321.
- [11] S. K. Zhou, *et al.*, “A review of deep learning in medical imaging: Imaging traits, technology trends, case studies with progress highlights, and future promises,” *arXIV:2008.09104*, 2020.

- [12] X. Chen, X. Wang, K. Zhang, *et al.*, “Recent advances and clinical applications of deep learning in medical image analysis,” *Medical Image Analysis*, 2022; 79:102444.
- [13] G. Litjens, *et al.*, “A survey on deep learning in medical image analysis,” *Medical Image Analysis*, 2017; 42:60–88.
- [14] M. A. Jones, R. Faiz, Y. Qiu, and B. Zheng, “Improving mammography lesion classification by optimal fusion of handcrafted and deep transfer learning features,” *Physics in Medicine and Biology*, 2022; 67(5):054001.
- [15] G. Danala, S.K.R. Maryada, W. Islam, *et al.*, “A Comparison of Computer-Aided Diagnosis Schemes Optimized Using Radiomics and Deep Transfer Learning Methods,” *Bioengineering*, 2022; 9(6):256.
- [16] V. Gulshan, *et al.*, “Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs,” *JAMA - Journal of the American Medical Association*, 2016; 316(22):2402–2410.
- [17] H. C. Shin, *et al.*, “Medical image synthesis for data augmentation and anonymization using generative adversarial networks,” *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 2018; 11037:1–11.
- [18] Y. Bar, I. Diamant, L. Wolf, *et al.*, “Chest pathology detection using deep learning with non-medical training,” *IEEE 12th International Symposium on Biomedical Imaging (ISBI)*, 2015; 294–297.
- [19] H. C. Shin, *et al.*, “Deep Convolutional Neural Networks for Computer-Aided Detection: CNN Architectures, Dataset Characteristics and Transfer Learning,” *IEEE Transactions on Medical Imaging*, 2016; 35(5):1285–1298.
- [20] J. Deng, W. Dong, R. Socher, *et al.*, “ImageNet: A large-scale hierarchical image database,” *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 2009; doi: 10.1109/cvpr.2009.5206848.
- [21] R. J. Chen, M. Y. Lu, *et al.*, “Synthetic data in machine learning for medicine and healthcare,” *Nature Biomedical Engineering*, 2021; 5(6):493–497.
- [22] M. Frid-Adar, I. Diamant, E. Klang, *et al.*, “GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification,” *Neurocomputing*, 2018; 321:321–331.
- [23] F. Mahmood, *et al.*, “Deep Adversarial Training for Multi-Organ Nuclei Segmentation in Histopathology Images,” *IEEE Trans Med Imaging*, 2020; 39(11):3257–3267.

- [24] F. Mahmood, R. Chen, and N. J. Durr, "Unsupervised Reverse Domain Adaptation for Synthetic Medical Images via Adversarial Training," *IEEE Transactions on Medical Imaging*, 2018; 37(12):2572–2581.
- [25] A. Waheed, M. Goyal, D. Gupta, *et al.*, "CovidGAN: Data Augmentation Using Auxiliary Classifier GAN for Improved Covid-19 Detection," *IEEE Access*, 2022; 8:91916–91923.
- [26] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," Sep. 2014, [Online]. Available: <http://arxiv.org/abs/1409.1556>
- [27] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," Dec. 2015, [Online]. Available: <http://arxiv.org/abs/1512.03385>
- [28] C. Szegedy, V. Vanhoucke, S. Ioffe, *et al.*, "Rethinking the Inception Architecture for Computer Vision," Dec. 2015, [Online]. Available: <http://arxiv.org/abs/1512.00567>
- [29] P. Porwal, *et al.*, "IDRiD: diabetic retinopathy – segmentation and grading challenge," *Medical Image Analysis*, 2021; 59:101561.
- [30] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," *CoRR*, vol. abs/1512.03385, 2015, [Online]. Available: <http://arxiv.org/abs/1512.03385>.